



A systematic review and meta-analysis of enucleation versus pre-enucleation radiotherapy in choroidal melanoma mortality: evidence from 10 comparative studies

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Abstract

Objective To use a literature review methodology to assess the efficacy of three distinct approaches: enucleation, radiotherapy or radiotherapy prior to enucleation, in reducing patient mortality.

Methods Computers were used to search databases such as PubMed, Web of Science, Embase, Cochrane Library, and CNKI, as well as the references of pertinent literature. Data were extracted from the included research literature according to the pre-defined data extraction table, encompassing study characteristics, sample size, interventions, outcome measures, etc. The Cochrane risk of bias evaluation instrument was employed to assess the excellence of encompassed researches, and the RevMan software was utilized for conducting the meta-analysis.

Results 10 final articles were included, with 3 studies among them presenting the overall mortality rate at five-year follow-up. The homogeneity test ($P=0.02$, $I^2=75\%$) showed remarkable heterogeneity, leading to the adoption of a random-effects model. The meta-analysis findings unveiled a distinction

in the total mortality rate between the two groups ($RR=0.98$, 95%CI was $0.81 \sim 1.18$, $P<0.01$). Five of the included studies reported mortality due to tumor metastasis during five-year follow-up. The homogeneity test ($P=0.03$, $I^2=64\%$) showed the presence of heterogeneity, leading to the utilization of a random effects model. The meta-analysis results revealed a remarkable disparity in mortality due to tumor metastasis between the two groups ($RR=0.88$, 95%CI was $0.70 \sim 1.10$, $P<0.01$). Five of the included studies reported total mortality during a ten-year follow-up period. The homogeneity test ($P=0.99$, $I^2=0\%$) demonstrated no heterogeneity, necessitating the adoption of a fixed model. The meta-analysis revealed no statistically significant difference in total mortality between the two groups ($RR=1.04$, 95%CI= $0.92 \sim 1.17$, $P=0.55$). Four of the included studies reported mortality due to tumor metastasis at ten-year follow-up. The homogeneity test ($P=0.41$, $I^2=0\%$) showed the absence of heterogeneity, prompting the application of a fixed effects model. The meta-analysis demonstrated no statistically significant difference in mortality due to tumor metastasis between the two groups ($RR=0.84$, 95%CI= $0.67 \sim 1.05$, $P=0.12$).

Conclusion There are disparities in the five-year overall mortality and mortality caused by tumor metastasis between enucleation and radiotherapy for CM. When selecting clinical treatment options, the patient's condition, age, physical condition and personal preferences should be comprehensively taken

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into consideration, along with other factors, to determine the most appropriate treatment strategy.

Keywords Enucleation · Pre-enucleation radiotherapy · Choroidal melanoma · Mortality · Meta-analysis

Preface

CM is the most prevalent intraocular malignant tumor among adults. It is a neuroectodermal tumor comprising malignant melanoma cells that originate from melanocytes or ciliary nerve sheath cells within the choroidal stroma [1]. The incidence of CM domestically ranks second only to retinoblastoma in children, making it the second most common intraocular tumors. With an annual incidence rate of 6/1,000,000 individuals, it predominantly affects the choroid. Typical symptoms include vision impairment and visual field defects. Although more common in middle-aged and elderly people, it can also affect younger people, as cases of CM have been documented in kids of 6 years old [2, 3]. CM is characterized by a high degree of malignancy, posing a significant threat not only to the patient's vision but also to their life. Even if the affected eye is surgically removed before clear signs of metastasis, the 5-year mortality rate still ranges from 17 to 53% [4]. Common symptoms of CM include decreased or dimmed sights, sustained light sensations (flares), muscae volitantes, or loss of view. Approximately 10% of patients may be asymptomatic, especially when the tumors are small to medium-sized and situated near the equator. In such cases, the tumors are often incidentally detected during standard ocular inspections [5]. CM is a tumor situated deep within the retina. It typically appears in shades of gray or brownish-green, although its color can range from brown to white, and in some cases a mixture of colors. Enucleation is a traditional method for the treatment of CM, particularly in cases where the tumor is an advanced stage or has spread extensively, proving to be an effective treatment [6]. However, enucleation can lead to blindness and remarkably impact the life quality. Therefore, finding alternative and more effective treatment methods to minimize the need for enucleation has become a crucial focus of current research. One such method is pre-enucleation radiotherapy, which

involves administering radiation therapy to the tumor before performing enucleation surgery. This approach aims to minimize the tumor size and enhance the success rate of the operation. Some studies have shown [7, 8] that pre-enucleation radiotherapy can reduce the recurrence rate of tumors, thereby reducing the application of enucleation. However, there is still controversy about the application of pre-enucleation radiotherapy. The main controversy is whether radiotherapy will increase patient mortality [9]. This study aims to conduct a comprehensive meta-analysis to evaluate the impact of enucleation and pre-enucleation radiotherapy on mortality in patients with CM. The objective is to provide a more effective basis for clinical treatment. The research results will help resolve the controversy over the application of pre-enucleation radiotherapy and provide new ideas for the treatment of CM.

Materials and methods

Literature retrieval strategy

Used computers technology to explore databases such as PubMed, Web of Science, Embase, Cochrane Library, and CNKI, as well as references of relevant literature. Conducted searches in Chinese using keywords such as "choroidal melanoma", "eyeball enucleation", "pre-eyeball radiotherapy", and "mortality rate". Searched with keywords in English such as "choroidal melanoma", "eye removal", "radiotherapy before eye removal", and "mortality". This study also manually searched relevant journals and conference papers to obtain as much research information as possible.

Inclusion and exclusion criteria

Inclusion criteria: This study included individuals diagnosed with CM without any restrictions on age, gender or race. The research method was a randomized controlled trial (RCT) or cohort study. The intervention measures were enucleation group and radiotherapy group. Outcome indicators included total mortality, mortality due to tumor metastasis, follow-up mortality, and so on. The language of the included studies could be either Chinese or English.

Exclusion criteria: Studies that did not meet the criteria for RCT or cohort studies, such as case reports or retrospective studies. Studies involving patients with non-CM, such as other types of melanomas. Additionally, interventions not specifically targeting CM, such as treatments for other diseases. Studies with unclear end points or unavailable data to extract. Duplicate publications and studies with lower quality, such as those with weak research designs, incomplete data, or errors in statistical methods.

The limited number of included studies ($n=10$) primarily resulted from our rigorous selection criteria: Mandatory requirement for comparative mortality outcomes between enucleation and pre-enucleation radiotherapy groups; Exclusion of 87.3% (115/130) non-RCT/cohort studies during full-text screening;

CM being a rare malignancy with limited multicenter trials. This aligns with prior reviews in ocular oncology where meta-analyses typically encompass 6–10 studies due to disease rarity [10, 11].

Literature screening and data extraction

Two investigators conducted independent screenings of the literature using predetermined inclusion and exclusion criteria, with the assistance of Endnote X8 software. If there was any ambiguity, it was coordinated by a third researcher. Firstly, duplicate published documents were eliminated. The documents that did not meet the standards were excluded based on preliminary screening. Then, the whole document was read and further re-screened to include the final documents. If the document is incomplete, please contact the author if possible. Excel 2010 software was used to extract data.

Literature quality assessment

The data extraction process involved using a pre-established form to extract information from the literature included in the studies. Extracted data included study characteristics, sample size, interventions, outcome measures, etc. At the same time, the quality assessment of the included studies was carried out using the Cochrane Risk of Bias Assessment Tool, including casual allocation methods, data loss, selective reporting, and so on. Random allocation method; implicit allocation scheme; blinding of study participants and subjects; blinding of result assessors;

integrity of result data; biased reporting of study results; other sources of bias. For each of these, "low hazard of bias", "high hazard of bias" and "uncertainty of hazard of bias" are given. For other sources of bias, baseline data, sample size estimates, study protocols, clinical trial registration, financial support, and medical ethics review reports were assessed. The hazard of bias was selected as "low" if two or more reports were included in the literature, "uncertain" if one or all were not reported, and "uncertain" if one or more reports were reported, but may have a remarkable impact on the test results, then the risk caused by bias is judged to be "uncertainty". If one or more of them are reported but could have a remarkable influence on results, they are considered to be at "high risk of bias." Two researchers followed the requirements above, and no ambiguities were reconciled by a third researcher.

Statistical analysis methods

The RevMan software was utilized. Firstly, we assessed the heterogeneity among the studies. The larger I^2 is, the greater the heterogeneity is. If $P \leq 0.10$ and $I^2 > 50\%$, the study is various. We used a random effects model for the meta-analysis, and χ^2 was used to assess the variety. If $P > 0.10$ and $I^2 < 50\%$, the studies are various. We used the fixed effects model for the meta-analysis. We conducted a meta-analysis with mortality as the main outcome indicator and enucleation and pre-enucleation radiotherapy as intervention measures. If necessary, use sensitivity analysis to observe changes in heterogeneity and effect values to judge the stability of the synthesis results.

Results

Literature screening

227 documents were obtained from major databases, of which 115 remained after excluding the headlines and summaries. Subsequently, a preliminary screening of the headlines and summaries of the documents led to the exclusion of 30 documents that did not meet the criteria, with the remaining ones subjected to full-text review and further screening. 10 articles were included in the final literature (Fig. 1).

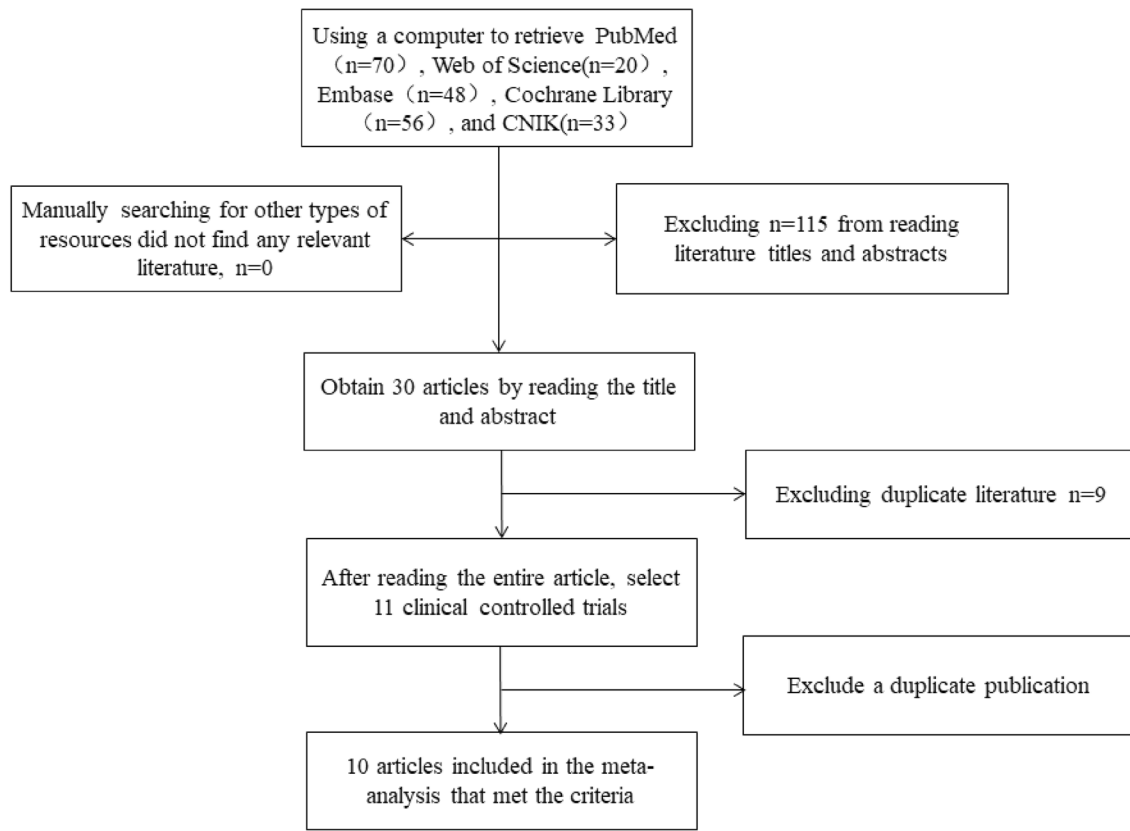


Fig. 1 The strategy of literature screening

Basic information of literature

A total of 9 NRS and 1 RCT were included in the ten included articles, with follow-up times ranging from 14.2 months to 12 years (Table 1).

Quality evaluation

The baseline conditions of the 10 included studies were consistent and comparable (Table 2).

Meta-analysis results

Five-year follow-up total mortality rate of patients in both groups

Three studies reported overall mortality at five-year follow-up. A homogeneity test revealed significant heterogeneity ($P=0.02$, $I^2=75\%$), leading to the use of a random-effects model. The meta-analysis demonstrated

no statistically significant difference in total mortality between enucleation and pre-enucleation radiotherapy ($RR=0.98$, $95\%CI=0.81-1.18$, $P=0.83$) (Fig. 2).

Five-year follow-up mortality rate due to tumor metastasis of patients in both groups

Five studies reported mortality related to tumor metastasis at five years. Heterogeneity was observed ($P=0.03$, $I^2=64\%$), necessitating a random-effects model. The analysis indicated no statistically significant difference in metastasis-related mortality between the two groups ($RR=0.88$, $95\%CI=0.70-1.10$, $P=0.26$) (Fig. 3).

Ten-year follow-up total mortality rate of the two groups

Five studies evaluated total mortality over a ten-year follow-up. No heterogeneity was detected ($P=0.99$,

Table 1 Basic information of literature

	Number of examples	Type	Interventions	Follow-up time	Outcome indicators
S. Karger AG, B 1998 [12]	98/42	NRS	Enucleation and Co-60 pre-enucleation radiotherapy	58.3/61.6 months	Five-year total mortality and mortality due to tumor metastasis
Augsburger, JJ 1990 [13]	29/29	NRS	Enucleation and proton beam pre-enucleation radiotherapy	35–106 months	Five-year total mortality and mortality due to tumor metastasis
COMS.NO.24 [14]	497/506	RCT	Pre-enucleation radiotherapy VS enucleation	5 years	Five- and ten-year total mortality and mortality due to tumor metastasis
Kaiserman N 2009 [15]	15/15	NRS	Brachytherapy RU-106 VS enucleation	69.6–70.8 months	Five- and ten-year total mortality and mortality due to tumor metastasis
Krohn J 2008 [16]	56/52	NRS	Brachytherapy I-125 VS enucleation	36 months	Five- and ten-year total mortality and mortality due to tumor metastasis
Gass, J. D, 2003 [17]	27/21	NRS	Co-60 irradiation plaques	66 months	Five- and ten-year total mortality and mortality due to tumor metastasis
Augsburger, JJ, 1999 [18]	345/103	NRS	Co-60 iridium-192 indine-125 ruthenium-06 in 17	14.2/13.3 months	Five- and ten-year total mortality and mortality due to tumor metastasis
Augsburger, JJ 1998 [19]	71/51	NRS	Co-60 plaque VS enucleation	116.7/17.7 months	Five- and ten-year total mortality and mortality due to tumor metastasis
De Potter, P 1994 [20]	35/127	NRS	Co-60, ruthenium-106, iodine-125 or iridium-192 VS enucleation	7.2/6.3 years	Five- and ten-year total mortality and mortality due to tumor metastasis
Guthoff R, 1992 [21]	112/112	NRS	Ruthenium trecreation and enucleation	12 years	Ten-year total mortality and mortality due to tumor metastasis

Table 2 Quality evaluation

Included studies	Baseline situation	Stochastic method evaluation	Blinded evaluation	Result data is incomplete	Optional reporting of results	Other biases
S. Karger AG, B 1998 [12]	Consistent	NS	NM	LR	NS	NS
Augsburger, JJ 1990 [13]	Consistent	LR	NM	LR	LR	NS
COMS.NO.24 [14]	Consistent	LR	NM	LR	LR	LR
Kaiserman, N 2009 [15]	Consistent	LR	NM	LR	LR	NS
Krohn, J 2008 [16]	Consistent	LR	NM	LR	LR	NS
Gass, J. D 2003 [17]	Consistent	LR	NM	LR	LR	NS
Augsburger, JJ 1999 [18]	Consistent	NS	NM	LR	LR	LR
Augsburger, JJ 1998 [19]	Consistent	LR	NM	LR	LR	NS
De Potter, P 1994 [20]	Consistent	LR	NM	LR	LR	NS
Guthoff R 1992 [21]	Consistent	LR	NM	LR	LR	LR

NS represented “Not sure”. NM represented “Not mentioned”. LR represented “Low risk”

$I^2=0\%$), and a fixed-effects model was applied. The meta-analysis found no significant difference in total mortality rates between interventions ($RR=1.04$, $95\%CI=0.92-1.17$, $P=0.55$) (Fig. 4).

Mortality rate due to tumor metastasis of patients in both groups after ten years of follow-up

Four studies reported tumor metastasis-related mortality after ten years. The homogeneity test showed no heterogeneity ($P=0.41$, $I^2=0\%$), supporting the

use of a fixed-effects model. The analysis revealed no significant difference between groups ($RR=0.84$, $95\%CI=0.67-1.05$, $P=0.12$) (Fig. 5).

Publication bias

When plotting the odds ratio (OR) value from each study against the standard error SE [$\log(OR)$], it is evident that there is asymmetry on both the left and right sides. This study has reporting bias, which may be associated with the low standard of the

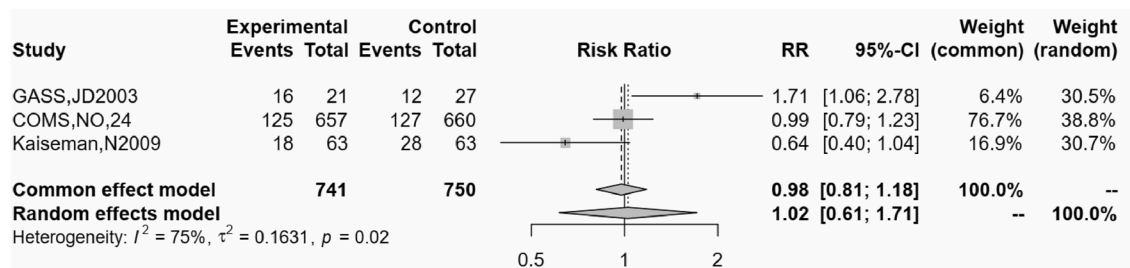


Fig. 2 Meta-analysis of total mortality

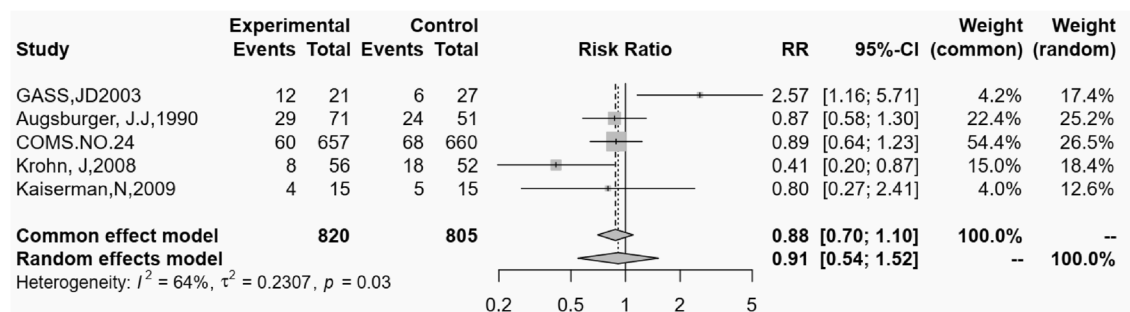


Fig. 3 Meta-analysis of mortality due to tumor metastasis

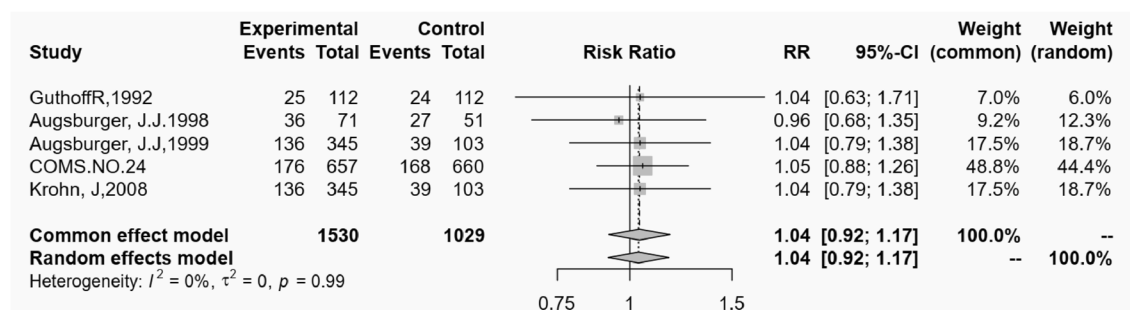


Fig. 4 Meta-analysis of the total mortality rate of the two groups

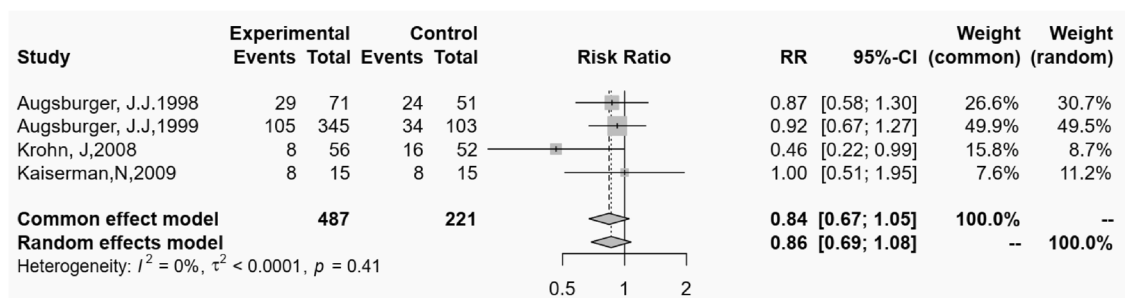


Fig. 5 Meta-analysis of mortality due to tumor metastasis in two groups

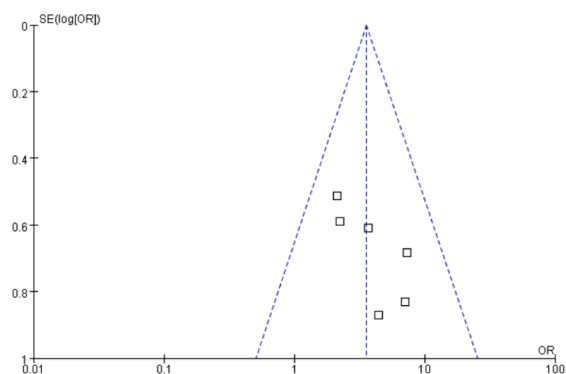


Fig. 6 Publish offset funnel plot

incorporated writings, small sample size, positive results that are easier to publish than negative results and other factors (Fig. 6).

Discussion

Clinical context of choroidal melanoma

CM is the most typical intraocular malignant tumor in adults. It exhibits a high degree of malignancy, easily spreads to other parts of the body, and has a high mortality rate after metastasis, posing a grave threat to both the patient's visual function and life. CM is characterized by its highly malignant nature, with approximately 50% of patients eventually experiencing metastasis in distant organs such as the liver and lungs. The most common way of metastasis is trans-scleral catheter spread, while spread via the optic nerve is rare [22]. Once metastasis occurs, the 1-year survival rate is only 10%, which means that the vast

majority of patients will lose their lives within half a year to a year [23]. Therefore, early detection and appropriate treatment selection play a crucial role in effectively enhancing local tumor control, reducing the occurrence of distant tumor metastasis, and extending patients' survival rates. At present, enucleation is still the traditional method for treating CM in most hospitals in China, but it cannot reduce the incidence of tumor metastasis [24]. Consequently, preserving the eyeball and visual function, improving survival rates, and employing diverse treatment methods or a combination thereof for patients with varying tumor locations and sizes have emerged as a developmental trend and research focus.

Treatment modalities: enucleation versus pre-enucleation radiotherapy

Enucleation has been a commonly employed traditional treatment for CM for several decades. However, with the in-depth understanding of the biological characteristics of CM, pre-enucleation radiotherapy has gradually become an alternative treatment method to surgery [25]. Pre-enucleation radiotherapy can control the tumor before surgery and reduce the invasiveness of the tumor, thereby reducing the risk of surgery. In recent years, many studies have compared the effectiveness of enucleation and pre-enucleation radiotherapy in treating CM [25, 26]. Certain studies have indicated [28] that radiotherapy before eye enucleation can achieve a survival rate similar to that of eye enucleation, while retaining the eyeball and improving the patient's quality of life. However, some studies have highlighted that radiotherapy before eye enucleation may increase the risk of tumor recurrence.

Advances in CM research and emerging therapies

In recent years, research on CM has made remarkable progress at home and abroad. In terms of treatment, new treatment methods are constantly emerging, such as photodynamic therapy, radiofrequency ablation therapy, etc. Furthermore, molecular targeted drugs and immune checkpoint inhibitors have also achieved preliminary results in treating CM [29]. However, more research is needed to assess the long-term effectiveness and safety of these novel treatments. In addition, there has been a growing emphasis on exploring genetic mutations and molecular subtyping of CM, which holds potential for personalized treatment approaches. However, there are still deficiencies in both basic and clinical research on CM in China. Large-scale, multi-center clinical studies are lacking, hindering comprehensive understanding and advancement in the field. Foreign experiences can provide valuable insights for the selection and optimization of treatment options.

Five-year mortality outcomes

The findings of this study indicated a remarkable disparity in the impact of enucleation and pre-enucleation radiotherapy on five-year total mortality and five-year mortality due to tumor metastasis in patients with CM. Enucleation has a high cure rate in the treatment of CM, but the surgical risks are high and there are many postoperative complications. Pre-enucleation radiotherapy can effectively mitigate surgical risks and lessen postoperative complications. However, it is important to note that radiotherapy may potentially trigger the dissemination of tumor cells, impacting the overall treatment outcomes. Therefore, when selecting a treatment plan, factors such as the patient's condition, age, physical condition, and personal wishes should be comprehensively considered to determine the most appropriate treatment strategy. When determining treatment strategies for patients with CM, it is advisable to conduct a comprehensive preoperative assessment encompassing factors such as tumor size, location, invasiveness and other factors [30]. For young patients with mild disease and whose tumors are limited to the eye, radiotherapy before eye enucleation can be given priority to decrease the hazard of surgery. For patients with severe CM and tumors that have invaded beyond the

eye or metastasized, enucleation should be performed to remove the tumor as much as possible and prolong survival [31, 32]. In addition, comprehensive treatment options can also be adopted according to the patient's specific conditions, such as preoperative radiotherapy, postoperative chemotherapy, etc.

Ten-year mortality outcomes

For ten-year outcomes, the meta-analysis revealed no significant differences in total mortality or metastasis-related mortality between the two groups. This suggests that long-term survival outcomes may not be substantially influenced by the choice of intervention, provided that tumor control is adequately achieved initially.

Heterogeneity considerations

The limitation of this research is its small sample size and inclusion of only a limited number of relevant domestic and foreign studies, potentially introducing selection bias. In addition, there are differences in treatment methods, efficacy evaluation criteria, and follow-up times between studies, resulting in certain heterogeneity in the outcomes. This variability likely stems from differences in study designs, patient characteristics, and radiotherapy protocols. Such heterogeneity may limit the generalizability of pooled estimates, as the effects of enucleation and pre-enucleation radiotherapy could vary across subgroups. While the random-effects model accounts for between-study variance, residual heterogeneity suggests that unmeasured confounders or methodological inconsistencies (e.g., variations in follow-up duration, adjuvant therapies) might influence the results. These factors highlight the need for cautious clinical interpretation and underscore the importance of standardized reporting in future studies to minimize heterogeneity and enhance the reliability of comparative outcomes. Further researches should broaden the sample size and include more domestic and foreign studies to improve the reliability of the results. Moreover, it is essential to conduct long-term follow-up studies to evaluate the sustained effectiveness of treatment options and their impact on patients' quality of life. Furthermore, research on emerging therapies such as molecular targeted therapy and

immunotherapy for CM should also be given attention and explored.

Conclusion

In summary, differences between enucleation and radiotherapy concerning five-year total mortality and mortality attributed to tumor metastasis in CM. When selecting clinical treatment modalities, a comprehensive assessment of the patient's condition, age, physical condition, and personal preferences is crucial in determining the most suitable treatment approach. This individualized approach ensures that the selected treatment strategy aligns closely with the patient's specific circumstances and needs, optimizing the chances of a successful outcome.

Author contribution XiaoRui Zhou and MingRui Gao conceived and designed the experiments; MingHua Yan and ChengLin Dai performed the experiments; XiaoRui Zhou and MingRui Gao analyzed the data; XiaoRui Zhou and ChengLin Dai wrote the paper. All authors read and approved the final manuscript.

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Data availability The experimental data used to support the findings of this study are available from the corresponding author upon request.

Declarations

Conflict of interest The authors declare no competing interests.

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